

Social physique anxiety experiences in physical activity: a comprehensive synthesis of research studies focused on measurement, theory, and predictors and outcomes

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Physical activity (PA) settings such as sport, exercise, and physical education are inherently social and evaluative in nature, with much attention directed at how the body looks and performs. As such, these settings foster a host of positive and negative emotional experiences. Social physique anxiety (SPA) is a commonly studied emotion that emanates from, or motivates, PA behaviors. In this synthesis of the 126 identified research papers focused on SPA and related experiences in sport, exercise, and physical education, we summarize the theoretical roots and measurement of SPA, present research evidence on predictors and outcomes of SPA experiences, explore links between SPA and PA behaviors, and offer insight on the future directions of SPA research. Specifically, researchers in sport and exercise psychology are encouraged to broaden the study of SPA to conduct longitudinal and experimental studies, develop quality intervention strategies aimed at reducing SPA experiences, and expand on the theoretical and operational understandings of SPA experiences across the lifespan.

Keywords: social physique anxiety; physical activity; review

Introduction

Physical activity (PA) settings such as sport, exercise, and physical education are inherently social and evaluative in nature, and much emphasis is placed on the body form and function. As such, these settings foster a host of positive and negative emotional experiences. With roots in social anxiety, self-presentation, and body image affect, social physique anxiety (SPA) is a commonly studied emotion that emanates from, or motivates, PA behaviors. SPA is defined as the anxiety that an individual experiences when he or she perceives that others could be negatively evaluating his or her physique (Hart, Leary, & Rejeski, 1989). Crawford and Eklund (1994) further describe SPA as self-presentational anxiety associated with the physique (e.g., body fat, muscularity, tone, body proportions). Given these operationalizations, it is not surprising that SPA experiences have been related to PA participation motives, attitudes, preferences, perceptions of self and ability, and engagement in or avoidance of PA. Nonetheless, one of the challenges in this area of research is to understand how SPA predictors and outcomes are interconnected and whether frameworks can be developed to better understand SPA experiences.

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SPA has theoretical roots in both body image and body esteem literatures and bridges one's image of his/her physique with the satisfaction or dissatisfaction of this image (Hart et al., 1989). With body image defined as a multidimensional construct consisting of perceptual, cognitive, affective, and behavioral domains (Cash & Smolak, 2011), SPA is often studied as one aspect of the affective dimension (Bane & McAuley, 1998). Furthermore, SPA introduces an interpersonal aspect of body esteem since it concerns how individuals feel about others' perceptions of their bodies. This latter perspective is rooted in self-presentation theory whereby individuals are motivated to make good impressions on others for social and material gains (Schlenker & Leary, 1982). Specifically, self-presentation is a goal-directed attempt to control how the self is perceived by an actual or perceived audience (Schlenker, 1980) and is a complex process involving both an individual's motivation to make a desirable impression and perceived ability to construct a desirable impression for others (Leary & Kowalski, 1990; Martin Ginis & Leary, 2004). Particularly in Western cultures, these 'desirable' impressions are often focused on attractive appearance and physical characteristics such as a thin and toned body shape for females and a muscular toned physique for males (Thompson, Heinberg, Altabe, & Tantleff-Dunn, 1999). When positive or favorable impressions are not possible or likely, this leads to affective experiences of social anxiety (Schlenker & Leary, 1982). Social *physique* anxiety experiences are a result of individuals being (or perceiving themselves to be) unable to make favorable impressions targeting their physical attributes such as body size and appearance (Hart et al., 1989). Based on self-presentation perspectives, individuals may choose to either engage in or avoid PA in order to improve their chances of making positive impressions; or perhaps to avoid circumstances in which the physique could potentially be evaluated negatively by others (Crawford & Eklund, 1994; Hart et al., 1989). This is the argument that is often made when researchers find a positive *or* negative association between SPA and PA. In fact, SPA is often used as a proxy of self-presentational processes (see Martin Ginis & Mack, 2012) or body image affect (Bane & McAuley, 1998).

Aside from the theoretical roots within self-presentation, social anxiety, and body esteem research, there are no specific conceptual or theoretical models related to SPA experiences in sport and exercise psychology. Nonetheless, SPA has been studied with concepts of motivation, personality, self-esteem, social relationships, emotion, and/or health behaviors (i.e. Anshel & Seipel, 2007; Cox, Ullrich-French, Madonia, & Witty 2011; Cumming & Duda, 2012; Frederick & Morrison, 1996; Haase, Mountford, & Waller, 2007; Hagger, Hein, & Chatzisarantis, 2011; Hagger & Stevenson, 2010; Koyuncu, Tok, Canpolat, & Catikkas, 2010; Magnus, Kowlaksi, & McHugh, 2010; Smith, Wright, & Winrow, 2010; Yin & Ryska, 1999). As such, it may be helpful to understand nomological networks or interactions between these related constructs. A nomological network is defined as a system of related constructs, their observable manifestations, and the interrelationships among these constructs (Cronbach & Meehl, 1955). To date, there are no comprehensive reviews that can be used to understand the complex interrelationships between predictors and outcomes of SPA experiences for advancing the field of sport and exercise psychology.

The purpose of the current paper was to comprehensively synthesize the research on SPA in PA contexts including exercise, sport, and physical education. Specifically, the objectives included: (1) examining measurement issues and construct properties associated with SPA; (2) identifying predictors and outcomes of SPA experiences; and (3) exploring nomological networks that have integrated SPA and PA.

To provide some context for the current synthesis, there is currently only one published review specific to SPA experiences (Mack, Wilson, Waddell, & Gasparotto, 2008). This meta-analysis consists of an overview of 16 studies published before 2007 that examined the empirical effect of SPA in PA contexts. Due to the specific inclusion criteria and requirements for a meta-analysis, this existing review offers a limited insight on the complexity of SPA experiences in sport and exercise psychology. As such, the present review focuses on providing a much broader, comprehensive and integrated synthesis of SPA research across a variety of contexts that are important in sport and exercise psychology (see Supplemental Table 1). Specifically, this comprehensive review incorporates all located original research examining theoretical roots and measurement of SPA, as well as evidence on predictors and outcomes of SPA experiences from 126 studies that have been published up to the summer of 2013. All research studies included in this comprehensive review are identified in Supplemental Table 2 and organized generally by PA (exercise, sport, physical education), health (social relationships, body image, eating/dieting), or measurement contexts.

Methods

Information retrieval

In this comprehensive review, White's (1994) guidelines were followed such that inclusion and search criteria were established, and multiple methods were used for information retrieval. Studies had to be (1) published in English; (2) qualitative or quantitative original studies published in a peer-reviewed journal (i.e., not a review, meta-analysis, commentary, book chapter, or supplement abstract), (3) involve a sport, exercise, physical education, or health context; (4) if quantitative design, the use of a version of the Social Physique Anxiety Scale (SPAS) (Hart et al., 1989) was essential and if qualitative design, the definition of SPA as reflective of Hart et al. (1989) and/or Crawford and Eklund (1994)'s operationalization was needed. Studies also had to be focused on human subjects, with a sample size of one or greater. Studies that examined a similar construct to SPA (i.e. self-presentation, body image), and not SPA directly, were excluded from the current review. Computer literature searches included an extensive exploration of various online databases spanning the year the SPA construct was operationalized (1989) until March 2013: ERIC, MEDLINE, PsycINFO, PubMed, Scopus, SPORTDISCUS, and Web of Knowledge. Key search terms were 'social physique anxiety', 'social physique', 'SPA', 'physique anxiety' AND 'PA', 'physical education', 'exercise', 'sport', and/or 'health'. An ancestry analysis was also conducted manually in which references were obtained from the bibliographies of articles retrieved through computerized literature searches and book chapters. Finally, some articles were retrieved by informal consultation with expert colleagues in the field and contacts with authors of primary studies identified in the computer search. After final article retrieval, the second author organized the appropriate articles by the research design, context, and measurement scale reported in each article. The number of articles and percentages per category are summarized in Supplemental Table 1. Brief summaries of each article were written to highlight the purpose, design, targeted sample and size, SPA measure utilized, correlates tested, and main study findings. The first author then confirmed this classification of articles by reviewing all articles, summaries, and coded information. Based on the general domain of focus and sample descriptives, each article/summary was then classified in one of the following contexts: (1) exercise, (2) sport, (3) physical education, (4) measurement, (5)

physique-related and health (Supplemental Table 2). For example, studies were classified in a measurement context if they tested the properties of any version of the SPAS. Similarly, studies were classified in a physique-related and health context if they focused on main health-related correlates such as disordered eating or depression or if the research was conceptualized in a physique-related domain that was not exercise, sport, or physical education. Classification of articles in Supplemental Table 2 underwent the same review process by the first author as outlined above.

Results

One hundred and twenty-six articles were identified, of which $n = 123$ (97.6%) were of quantitative design and $n = 3$ (2.4%) described using qualitative methodologies. The majority of the 126 articles ($n = 89$; 70%) were cross-sectional designs compared to seven (6%) longitudinal and 27 (21%) experimental or intervention designs. Of interest, only one review article (Mack et al., 2008) and one commentary (Eklund, 1998) were located and used to inform the literature search but not included in the overall synthesis. Among the identified and incorporated articles, there were 64 (51%) that examined SPA in an exercise context, 17 (13.5%) framed within a sport context, 5 (4%) in a physical education context, and 27 (21.4%) in other health-related contexts such as body image and eating/dieting, and 13 (10%) in measurement studies. Supplemental Table 1 provides an overview of studies focused on SPA. All articles in this synthesis of SPA research are presented in Supplemental Table 2, organized by the general context of focus and with information on the purpose, study design, target sample and size, SPA measure used, tested correlates, and main findings provided for each article. In a provisional effort to illustrate the nomological networks relating to SPA, Figure 1 synthesizes the findings from this review. The main findings that emanate from the research on SPA are summarized in reference to the main study objectives: (1) to examine the main measure of SPA and state versus trait properties of the construct, (2) to identify predictors and outcomes of SPA experiences across the lifespan and (3) to explore the nomological networks that integrate SPA within PA behavior.

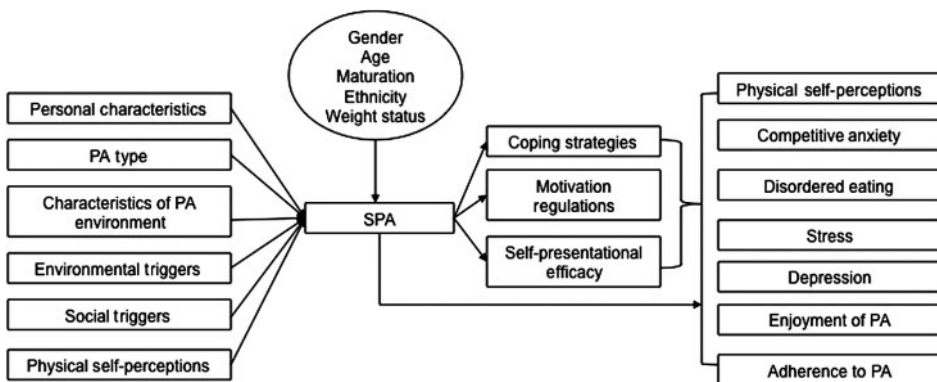


Figure 1. A synthesis of factors associated with experiences of social physique anxiety.

Measurement of SPA

Scale development and properties

The Social Physique Anxiety Scale (SPAS) was developed as a 12-item scale and tested for internal consistency, test-retest reliability, and both construct and criterion validity (Hart et al., 1989). SPA scores were moderately to highly correlated with a range of personality traits, anxiety, self-presentation processes, and body image dimensions. Early research exploring the measurement properties of the SPAS supported the use of the original 12-item scale, and offered insight into a two-factor scale comprised of physique presentation comfort (sample item 8: 'I am comfortable with how fit my body appears to others') and expectations of negative physique evaluation (sample item 9: 'I am uncomfortable to know others were evaluating my physique/figure'; Eklund, Kelley, & Wilson, 1997; Eklund, Mack, & Hart, 1996; Petrie, Diehl, Rogers, & Johnson, 1996). Further research argued for a reduced 9-item scale deleting three of the five positively worded items and maintaining a unidimensional structure (Martin, Rejeski, Leary, McAuley, & Bane, 1997). Of the three deleted items, item 2 ('I would never worry about wearing clothes that might make me look too thin or overweight') was deleted due to overall participant confusion. Items 1 ('I am comfortable with the appearance of my physique/figure') and 5 ('When I look in the mirror I feel good about my physique/figure') were removed and justified because they did not involve a social evaluative component (Martin et al., 1997).

Additional testing of the measurement scale has provided evidence for the 9-item or an 8-item unidimensional scale with the additional deletion of the item 12 (i.e., 'When in a bathing suit, I often feel nervous about the shape of my body'; Motl & Conroy, 2000). For younger adolescents, researchers have also advocated for a 7-item scale essentially removing all positively-worded items from the scale (Motl & Conroy, 2001; Smith, 2004). Even with reverse-coding protocols in place, most of the problems with the factor structure of the SPAS have emanated from the item valence, whereby high correlations among similarly valenced items (positive or negative) and weak correlations between positive and negative items are reported. Nonetheless, the majority (97%) of researchers report using the 9- or 12- item scale (Supplemental Table 1), with high internal consistency coefficients for the SPAS items (e.g., Cronbach's alpha coefficients are consistently reported to be higher than 0.85) across diverse samples.

Furthermore, there are a number of accounts of the SPAS being translated into different languages and tested for invariance across cultures, gender, and age (Hagger et al., 2007, 2010; Lindwall, 2004; Maïano et al., 2010). Generally, the underlying concept of SPA has been demonstrated as an important body image dimension across cultures.

In addition to the trait-SPA measure (SPAS), and based on supporting evidence and growing research interest in understanding experiences of state-SPA, Martin Ginis et al. (2011) developed a short ($N_{\text{items}} = 9$) state-SPA scale. Scores on the state-SPA scale provided evidence of construct validity (i.e., significant positive correlations with scores of trait and state measures of body image affect, and negative correlations with self-presentational efficacy and body satisfaction). Furthermore, state-SPA scores discriminated between females who exercised in a mixed-sex environment versus a same-sex environment whereas trait-SPA scores did not.

In summary, there are both trait and state measures of SPA available to researchers who are interested in understanding SPA experiences. It is imperative that the chosen

measure fits the research objectives and target sample in studies focused on developing this area of sport and exercise psychology science.

The trait versus state nature of SPA

The original conceptualization of SPA was as a stable disposition or trait (Hart et al., 1989). However, there are few longitudinal studies that have looked at the stability of SPA scores over time. One of these longitudinal studies provides strong evidence of SPA as a stable trait in a sample of women during a vulnerable time of fluctuating body concerns and social interactions. Specifically, Crocker et al. (2006) conducted a prospective longitudinal study with 501 adolescent girls aged 14 to 17 years, and demonstrated that scores on the SPAS were quite stable over three time waves one year apart (e.g., interclass correlation coefficients of $r_{\text{time1_time2}} = .69$, $r_{\text{time2_time3}} = .69$, and $r_{\text{time1_time3}} = .58$). The stability of SPA experiences is consistent with findings from a systematic review of longitudinal body image research, suggesting that body image concerns remain fairly stable in individuals between 18 and 80 years old (Davis, 1997). While these studies provide limited evidence, the trait-like features of SPA are supported.

Alternatively, several researchers have provided evidence of state-like properties of SPA scores both using the original scale and modified items to target situation-specific physique anxiety. Generally, SPA has demonstrated malleable features in physical, interpersonal, and intrapersonal contexts. Several researchers have explored differences in SPA based on the physical or intrapersonal contexts (Brunet & Sabiston, 2011; Carron, Burke, & Prapavessis, 2004; Chu, Bushman, & Woodard, 2008; Diehl et al., 2001; Focht & Hausenblas, 2001, 2003, 2006; Kruisselbrink, Dodge, Swanburg, & MacLeod, 2004; Lamarche & Gammage, 2010; Martin Ginis, Jung, & Gauvin, 2003; Sabiston, Sedgwick, Crocker, Kowalski, & Mack, 2007; Sinden, Martin Ginis, & Angove, 2003; Van Raalte, Cunningham, Cornelius, & Brewer, 2004). For example, Van Raalte et al. (2004) reported that SPAS scores were higher among college students reading hypothetical scenarios placing them in a fitness center when compared to a dining hall or library. These findings suggested that physical environments where the body may be on display and evaluated (e.g., fitness center) fostered the highest SPA experiences compared to more private settings such as the library. Similarly, exercising in front of a mirror – an environment that highlights that the body is on display – has been linked to higher reports of SPA (e.g., Gammage, Hall, & Martin Ginis, 2004; Martin Ginis, Jung, & Gauvin, 2003). In addition, women who exercised to a physique-salient exercise video (compared to a group who exercised to a non-physique salient video) reported higher state-SPA (Martin Ginis, Prapavessis, & Haase, 2008). Taken together, the noted differences in SPA scores resulting from hypothetical or actual manipulations to the physical environment may allude to state-like properties of SPA experiences. These findings also highlight important environmental factors that affect SPA experiences.

Taking an interpersonal context approach, Brunet and Sabiston (2011) reported higher SPA scores among older adolescents when they were in the company of peers compared to being around their parents. Furthermore, female fitness club members reported higher situational-specific SPA in all-male (i.e., opposite sex) exercise environments compared to mixed-sex and all-female settings in hypothetical scenarios (Kruisselbrink et al., 2004). Contrary to these findings, Lamarche, Gammage, and Strong (2009) reported no difference in SPA scores for individuals engaged in a male-led or female-led exercise class. Also, Lamarche, Gammage, and Gabriel (2011) reported that state-SPA did not

differ for men and women when randomly assigned to complete a fitness task with a same-sex or opposite-sex experimenter. Features of the social environment may therefore induce SPA experiences differently, giving merit to researching state-specific experiences of SPA. As such, state-SPA needs to be addressed both in sport and exercise psychology research and practice – for example, in counseling and mental training, coaching, and physical education.

In summary, the results provide evidence that SPA experiences can be conceptualized as either a stable or situation-specific variable. Researchers need to employ the dispositional conceptualization of SPA that best matches their sport and exercise psychology research goals.

Predictors of SPA

Research on SPA experiences in sport and exercise psychology has focused on identifying and classifying predictors of SPA. Predictors of SPA have included personal characteristics such as age, sex/gender, culture, and weight status, PA type, social triggers, and a range of self-perceptions. Unfortunately, the majority of the evidence is based on cross-sectional research designs, ultimately precluding directionality of effects.

Personal characteristics

The characteristics that have been predominantly examined as reflecting SPA score differences and moderator effects include age, sex/gender, weight status, and culture/ethnicity. There is cross-sectional evidence predominantly in the exercise context that SPA scores are lower among older adults compared to younger and middle-aged adults (Hausenblas & Martin, 2000; Lantz, Hardy, & Ainsworth, 1997; McAuley, Bane, Rudolph, & Lox, 1995; Ransdell, Wells, Manore, Swan, & Corbin, 1998; Treasure, Lox, & Lawton, 1998). This trend is true for males and females (Thøgersen-Ntoumani & Ntoumanis, 2006). On the contrary, some studies with older adults (Lanning, Bowden, Owens, & Massey-Stokes, 2004; Woodgate, Martin Ginis, & Sinden, 2003) and post-menopausal women (Ransdell et al., 1998) have found no association between age and SPA. Focusing specifically on adolescence, one longitudinal study within the exercise context focused on stages of maturation among girls. Findings indicated that SPA was positively correlated with maturation in adolescent girls such that girls in early stages of maturation reported significantly lower SPA than girls in middle or late stages of maturation (Niven, Fawkner, Knowles, Henretty, & Stephenson, 2009). The trend of increasing SPA with maturation in adolescence combined with research of higher SPA in young adult females compared to pre-adolescents (Monsma, Pfeiffer, & Malina, 2008; Thompson & Chad, 2002) suggests an important role of maturation and age in SPA experiences, especially in adolescence. Further research is needed to better understand the developmental features of the experience of SPA, in particular among males and during critical times of physical, emotional, and social change such as puberty.

Sex differences are commonplace among SPA research evidence in all contexts related to PA (i.e., 30.2% of the studies identified in Supplemental Table 2 are focused on sex differences). Specifically, there is consistent evidence that females score higher on the SPAS than males (e.g., Berry & Howe, 2004; Eklund et al., 1997; Mack, Strong, Kowalski, & Crocker, 2007) and these differences are reported across the lifespan (Melbye, Tenenbaum, & Eklund, 2008; Smith, 2004). Nonetheless, males are not protected from

experiencing SPA and related emotions (i.e., Crozier, 2012; Davis, Brewer, & Weinstein, 1993; Grieve et al., 2008). Furthermore, there is evidence that reported mean differences reflect true sex differences in SPA scores since multiple studies have demonstrated sex invariance in the measurement properties of the SPAS (i.e., Motl & Conroy, 2000, 2001; Smith, 2004).

Weight status has also been studied extensively as a correlate of SPAS scores predominantly within exercise and other health-related contexts such as eating/dieting behaviors and body image. Body mass index (BMI) has been consistently used as a measure of weight status when examining SPA (Crocker, Snyder, Kowalski, & Hoar, 2000; Krane, Stiles-Shipley, Waldron, & Michalenok, 2001). High BMI has been associated with higher SPA scores in adolescents (Crocker et al., 2003; Mack et al., 2007) and adults (Brewer, Diehl, Cornelius, Joshua, & Van Raalte, 2004; Russell, 2002). Overweight men and women tend to report higher SPAS scores compared to age and gender-matched healthy weight counterparts (Hart et al., 1989; Lantz et al., 1997; McAuley et al., 1995). Similar to BMI, body fat percentage (Cox, Lantz, & Mayhew, 1997; Martin, Kliber, Kulinna, & Fahlman, 2006; Ransdell et al., 1998), skinfold measurement (Monsma et al., 2008; Thompson & Chad, 2002), and dual-energy X-ray absorptiometry (DEXA; Martin Ginis, McEwan, Josse, & Phillips, 2012) have also been positively correlated with SPA. However, reports from a sample of male professionals in India suggest no significant association between anthropometrics and SPA (Mookerjee, Singh, & Cash, 2002). This inconsistent finding may be explained by cultural differences in weight perceptions and body image ideals. Researchers are encouraged to explore mechanisms such as cultural attitudes and beliefs about weight that explain the association between weight status and SPA experiences to help tease apart the inconsistent findings. Also related to weight status, perceptions of weight and muscle-related body comparisons are significantly tied to higher SPA scores among males and females (i.e., Martin Ginis et al., 2005; McCreary & Saucier, 2009).

There is limited research evidence exploring cultural differences in SPA experiences ($n_{studies} = 5$; 3.9%). Consistent with body image research more generally (Cash & Smolak, 2011), Caucasian/White participants have reported significantly higher SPAS scores compared to age-matched Black participants (Jordan, Smisson, Burke, Joyner, & Czech, 2005; Russell, 2002; Russell & Cox, 2003). In two studies exploring cross-cultural invariance, Hagger et al. (2007, 2010) reported some SPA measurement differences across participants from Spain, Portugal, Britain, Estonia, Sweden, and Turkey. For example, the 8-item SPAS exhibited good fit with British, Estonian, and Swedish samples while the 7-item version was a good fit with Spanish and Turkish samples (Hagger et al., 2007). They also reported that British and Spanish participants reported the highest SPA. Both the cultural differences and the measurement differences are preliminary evidence that requires further testing along with considerations for mechanisms explaining the relationship between culture/ethnicity and SPA experiences.

PA type and characteristics

Exercisers tend to report lower SPA experiences compared to inactive and non-exercising individuals (Crocker et al., 2000; Haase & Prapavessis, 2001; Hausenblas & Mack, 1999; Ransdell et al., 1998). Athletes in competitive aesthetic and subjectively rated sports such as figure skating, gymnastics, cheerleading, and some martial arts tend to report higher SPAS scores compared to athletes in non-aesthetic sports such as basketball and soccer.

This difference in SPAS scores is likely due to the high importance of aesthetics, lean body types, and the evaluative judging component of certain sports (Greenleaf, 2004; Hausenblas & Carron, 1999; Reel & Gill, 1996, 2001; Schwerin et al., 1996; Van Raalte, Schmelzer, Smith, & Brewer, 1998). Evidence is also less clear when comparing significant differences in SPAS scores among athletes compared to exercisers, or competitive versus recreational athletes. In one study of Turkish male and female adults, competitive athletes and exercisers had lower SPAS scores than non-exercisers, independent of gender (Mülazimoglu-Balli, Koca, & Aşçi, 2010). Haase and Prapavessis (2001) found no significant differences in SPAS scores comparing females with varying levels of sport experiences – including physique-salient sport athletes, weight-restricted sport athletes, non-physique-salient sport athletes and non-athletes. Significant differences on SPAS scores were reported among experienced and inexperienced bodybuilders and weightlifters, with the former reporting the lowest scores (Hurst, Hale, Smith, & Collins, 2000). Drawing from a meta-analytical review of SPA literature, there is a small overall consensus that exercisers report lower SPAS scores compared to non-exercisers, and SPAS scores may be higher among athletes compared to exercisers (Mack et al., 2008).

There have also been a few experimental studies providing evidence on the effects of PA type on SPAS scores. In a pre-post study design without controlling for body composition indicators, Bartlewski et al. (1996) found that SPAS scores were not different between women assigned to an aerobic exercise condition compared to a control condition. Bowden et al. (2005) reported an investigation of 315 undergraduate students enrolled in either a 16-week course for PA or an emergency care course. Participants enrolled in a PA course reported higher SPAS scores at baseline and greater improvements in SPA over the 16-week observation period. Changes in SPA scores did not appear halfway through the course, at eight weeks, suggesting that SPA can be attributed to dispositional beliefs that may take time to change. However, the researchers did not control for body composition or related measures that may help to explain the findings. Furthermore, Lindwall and Lindgren (2005) found that an exercise intervention significantly reduced SPA experiences among adolescent females compared to a control group, and these differences were not attributed to changes in physiological factors. In longer intervention studies, McAuley et al. (1995, 2002) reported significant reductions in SPAS scores over 20 and 52 weeks among older sedentary adults. In one of the aforementioned studies (McAuley et al., 1995), reductions in SPAS scores were more dominant for women than men and for younger (45–54 years) rather than older (55–64 years) participants following the exercise intervention. Finally, Martin Ginis and colleagues found that change in SPA was different for men and women following a strength training intervention (Martin Ginis, Eng, Arbour, Hartman, & Phillips, 2005). SPA was negatively correlated with perceptions of strength, muscularity, and body fat for men but only perception of body fat for women. In another 16-week diet and PA intervention study, Martin Ginis and colleagues reported that change in SPA was related to change in percentage of fat among overweight women (Martin Ginis et al., 2012). Taken together, it is important to tease out the relative (and cumulative) effects of PA type and characteristics of sport and exercise contexts on fostering or protecting from SPA experiences.

Environmental triggers

As reported in the section on state-SPA, there are a number of environmental predictors of SPA related to the characteristics of the environment (e.g., mirrors, same versus mixed

sex, attire restrictions). Exercising in front of mirrors has been shown to exacerbate SPA experiences in some research studies (Focht & Hausenblas, 2003; Katula, McAuley, Mihalko, & Bane, 1998; Martin Ginis et al., 2003) but not others (e.g., Raedeke, Focht, & Scales, 2007). Exercise attire may also exacerbate SPA experiences. In one of the original studies on this topic, Crawford and Eklund (1994) found women's SPA scores were negatively correlated with a preference for a physique-enhancing exercise environment (i.e., all exercisers in the video manipulation were wearing thong leotards and tights) and positively associated with favorable attitudes for exercising in an environment de-emphasizing the physique (i.e., all exercisers were wearing t-shirts and shorts). In sport settings, wearing physique-enhancing uniforms may draw attention to body shape, size, and musculature and increase chances of evaluation by coaches, teammates, athletic competitors, or spectators. Cheerleaders and swimmers have reported their revealing uniforms are a large source of pressure to maintain low body weight and exacerbate SPAS scores (Martin, 1999; Reel & Gill, 1996, 2001). However, in an examination of female aerobic exercisers and female varsity athletes whose uniforms ranged in level of revealing physique, Krane et al. (2001) found no difference in SPAS scores. Mandatory uniforms in physical education classes are also described as highly relevant sources of SPA among adolescents (Sabiston et al., 2007). A final environmental trigger of SPA experiences may be the gender make-up (i.e., mixed, same sex) of the PA environment (Kruisselbrink et al., 2004), although the evidence is mixed since SPAS scores were not significantly different among women attending same-gender or mixed-gender fitness facilities (Walton & Finkenberg, 2002) or when exercising with a same-sex or opposite-sex experimenter/trainer (Lamarche, Gammage, & Gabriel, 2011).

Social triggers

In addition to the personal and environmental characteristics that can influence experiences of SPA, there are numerous social contexts and related perceptions that appear to predict or distinguish SPAS scores. For example, peer group influences (Carron & Prapavessis, 1997) and the presence of others (Brunet & Sabiston, 2011; Spink, 1992) may perpetuate feelings of SPA. In a study by Mack et al. (2007), male and female adolescents who received encouragement from friends in their peer groups to alter their physiques, and who identified themselves as less attractive than their peers, reported higher SPAS scores. Cox et al. (2011) investigated several social relationship variables to determine which ones held the most relevance to students' SPA experiences in physical education. These included the presence of one's best friend in class, the number of close friends they had in class, perceived peer acceptance, and perceived teacher support. Of these relationship variables, only feelings of belonging or acceptance among one's peers were significant negative predictors of SPA in physical education. This suggests that having a good network of relationships with the broader peer group in class may attenuate SPA in the physical education setting, though not necessarily by having a few close friends or a supportive teacher.

Other interpersonal sources of SPA may include parents, teachers, coaches, fitness leaders, and the media. For example, mothers of adolescent girls may trigger SPA through indirect commentary and controlling actions such as guiding PA and managing their daughters' diets (i.e., Sabiston et al., 2007). Teachers and coaches who place an emphasis on appearance and weight are perceived to foster more threatening physique-salient and evaluative environments (Krane et al., 2001). Additionally, the media may heighten

experiences of SPA. In particular, there is evidence (predominantly studies involving women) that SPAS scores increase when individuals are exposed to appearance-salient advertising and decrease when individuals are exposed to health-based advertising (Berry & Howe, 2004). Similarly, Sabiston and Chandler (2009) found that SPA scores were higher for women exposed to model-focused athletic shoe advertising compared to women exposed to product-focused advertisements. These findings are consistent when the exposure includes fitness videos, television commercials, or print media such as photos and posters (Eklund & Crawford, 1994; Martin Ginis et al., 2008; Monro & Huon, 2005).

Self-perceptions

Body dissatisfaction and negative body image indicators have been consistent correlates of SPA experiences. For example, Thompson and Chad (2002) reported significant associations between body dissatisfaction, body weight and shape concerns, and SPAS scores. Negative correlations between appearance perceptions and SPAS scores have been reported (Amorose & Hollembeak, 2005; Crocker, Sabiston, Kowalski, McDonough, & Kowalski, 2006). A three-year longitudinal analysis reported decreases in physical self-perceptions (including sport, conditioning, and strength) associated with increases in SPA (Crocker et al., 2006). In a sample of female undergraduate students, Atalay and Gençöz (2008) reported that participants high on body dissatisfaction had higher SPA experiences. Self-esteem is also a significant predictor of SPAS scores among adolescents (Brunet, Sabiston, Dorsch, & McCreary, 2010) and undergraduate students (Lox, Osborn, & Pellett, 1998). Self-objectification has also been positively associated with SPAS scores (Melbye et al., 2008). Specific self-discrepancies (i.e., actual, ideal, and ought) have been significantly associated with heightened SPA experiences (Martin Ginis et al., 2008; Sabiston, Crocker, & Munroe-Chandler, 2005; Woodman & Steer, 2011). The findings that SPAS scores are highly related to body-related cognitions, perceptions, and affect are not surprising given that SPA is operationalized within a multidimensional body image framework (Bane & McAuley, 1998).

Outcomes of SPA

There are a range of additional outcomes that have been studied in the context of SPA experiences, such as physical self-perceptions, commitment to PA, competitive anxiety, disordered eating behaviors, stress, and depression. For example, SPA has been linked to low perceptions of competence (Brunet & Sabiston, 2009), self-efficacy (Marquez & McAuley, 2001), and low enjoyment (Cox et al., 2011; Raedeke et al., 2007, 2009) of PA. In a sample of male and female students with a sport history, SPA was associated with high competitive anxiety among the female athletes (Martin & Mack, 1996), supporting the theory that this anxiety may be influenced by fear of both performance and appearance being negatively evaluated. SPA has also been linked to PA commitment (Finkenberg, DiNucci, McCune, Chenette, & McCoy, 1998) and factors such as perceived evaluative threat, state anxiety, and self-presentational concerns during PA (Focht & Hausenblas, 2004; Gammage & Gabriel, 2009).

Researchers have also linked SPAS scores to disturbed eating attitudes and reports of disordered eating behaviors among college women and various elite sport athletes (e.g., Aşçi et al., 2006; Baş, Aşçi, Karabudak, & Kiziltan, 2004; Baş, Karabudak, & Kiziltan,

2005; Caglar et al., 2010; Cox et al., 1997; Diehl, Johnson, Rogers, & Petrie, 1998; Frederick & Morrison, 1998; Haase & Prapavessis, 1998, 2001; Haase, Prapavessis, & Owens, 2002; Hausenblas & Mack, 1999; Reel & Gill, 1996; Thompson & Chad, 2002). In addition to simple associative effects, mechanisms have been tested to explain the association between SPAS scores and eating-related outcomes. For example, Fitzsimmons-Craft, Harney, Brownstone, Higgins, and Bardone-Cone (2012) found that social comparison, appearance-related social comparison, and body surveillance significantly moderated the relationship between SPA and disordered eating in a sample of undergraduate females. Haase & Prapavessis (1998) reported that gender and body mass index interaction moderated the association between SPAS scores and eating attitudes. Sport type was not a significant moderator when examining disordered eating attitudes (Haase & Prapavessis, 2001).

Some recent reports have also associated physique evaluative threat, which may be analogous to SPA, to higher physiological markers of stress (i.e., Martin Ginis, Strong, Arent, & Bray, 2012). Specifically, individuals in contexts manipulating physique threat had high post-manipulation levels of cortisol, which is a hormone released by the hypothalamic-pituitary-adrenal axis. This study has demonstrated initial evidence that SPA may have long-term health implications since cortisol and heightened chronic stress are associated with a host of poor health outcomes (e.g., cardiovascular disease, immune system dysregulation, depression; Dickerson & Kemeny, 2004). Future research studies should focus on the overall long-term effects of SPA on psycho-biological markers and the possible link to poor health outcomes.

To this effect, there are five studies identifying SPA as positively associated with depression (Diehl, Johnston, Rogers, & Petrie, 1998; Lantz, Hardy, & Ainsworth, 1997; Lox, Osborn, & Pellet, 1998; Martin Ginis et al., 2012; Woodman & Steer, 2011). However, due to the cross-sectional nature of the findings, it is not possible to rule out the possibility that depression also leads to increases in SPA. To better understand the directionality of this relationship, longitudinal research and a focus on clinical samples are needed to further examine the mechanisms linking SPA and depression (as well as other mental health outcomes such as anxiety, stress, and mood).

Nomological networks: SPA and PA

There is certain ambiguity in the understanding of whether higher SPAS scores are related to participation in, or avoidance of, PA. Given the self-presentation and social anxiety foundations of SPA, it is not surprising that a predominant strategy for managing physique concerns may be behavioral avoidance (i.e., avoiding the situation or event that is likely to elicit the feelings of anxiety). For example, researchers have reported negative associations between SPAS scores and PA participation (Brunet & Sabiston, 2009; Kowalski, Crocker, & Kowalski, 2001; Lantz et al., 1997), and higher SPAS scores were associated with lower exercise adherence in a study of sedentary and obese females (Treasure et al., 1998). Similar associations can be seen in contexts of physical education (Cox et al., 2011), with SPA related to lower effort exerted, and directly related to participation avoidance in a high school class. In another sample of physical education students, there were no differences in SPA among those in mixed-sex and single-sex physical education classes, although females reported a preference for single-sex physical education classes (Koca & Aşçi, 2006). There is also qualitative evidence that adolescents who experience high SPA in physical education classes are more likely to avoid classes

by skipping physical education altogether or seeking parental notes excusing them from class (e.g., Sabiston et al., 2007).

When avoidance of the sport and/or exercise situation is not feasible, there are reports that individuals high in SPA will manage the threatening context by covering up their physique to avoid potential evaluations by others. For example, adolescent girls reported wearing loose-fitting clothing in physical education classes and at the gym, and a common strategy at the beach or swimming pool is to cover a bathing suit and avoid exposing the body unless necessary (Kowalski, Mack, Crocker, Niefer, & Fleming, 2006; Sabiston et al., 2007). These behavioral avoidance and remedial actions to alter one's physique or appearance (and hence lower SPA experiences) are consistent with two prominent ways in which individuals manage their SPA experiences (Hart et al., 1989; Kowalski et al., 2006; Sabiston et al., 2007).

While some researchers have identified negative associations between SPAS scores and PA, there is also evidence that SPA experiences are unrelated to PA behaviors (e.g., Cox et al., 2011; Crawford & Eklund, 1994; Melbye et al., 2008), or positively associated with engagement in PA (Aşçi, Tüzün, & Koca, 2006; Bowden, Rust, Dunsmore, & Briggs, 2005; Frederick & Morrison, 1996; Hausenblas, Brewer, & Van Raalte, 2004). In an attempt to explain these ambiguous findings, researchers have linked SPA experiences to reasons for exercise (Sabiston et al., 2005) and specific behavioral regulations, with a predominant focus on Deci and Ryan's (2002) self-determination theory perspectives (Brunet & Sabiston, 2010; Cox et al., 2011, 2013; Frederick & Morrison, 1996; Niven et al., 2009; Sabiston et al., 2005; Thøgersen-Ntoumani & Ntoumanis, 2006). Higher scores on the SPAS have been consistently associated with more controlling reasons for exercise, such as exercising to gain some reward or to avoid feelings of guilt and shame (e.g., Sabiston et al., 2005; Thøgersen-Ntoumani & Ntoumanis, 2006). More recently, exercise motives have been tested as mediators of the association between SPA experiences and PA behavior (Brunet & Sabiston, 2009) or physical education outcomes (Cox et al., 2011). In their study with older adolescents, Brunet and Sabiston (2009) tested both psychological needs (perceptions of relatedness, competence, and autonomy) and motivation as mediators of the SPA and PA association in a structural model. Among males and females, SPAS scores were negatively related to the perceptions of competence, autonomy, and relatedness, with competence predicting self-determined motivation and PA behavior. The model provided some evidence for the mediation of the needs and motivation as an explanation for the association between SPA experiences and PA.

In physical education settings, Cox et al. (2011) tested autonomous motivation, introjected and external regulations, and amotivation as unique mediators in the relationship between SPA and behavioral outcomes such as self-reported effort and avoidance of participation in physical education. There were no significant mediation effects, although SPA correlated negatively with autonomous motivation and positively with controlling forms of motivation and amotivation. Therefore, the type of motivation students experienced in physical education did not appear to explain the association between SPA experiences and certain behavioral outcomes. Given these findings, Cox et al. (2013) proceeded to take a person-centered approach to examining SPA experiences and behavioral regulations (using cluster analysis), and associations to relevant PA outcomes specific to physical education. Difference tests showed that the 'externally controlled and anxious students' reported significantly lower enjoyment and effort in class than the other identified profiles. As such, it appears that SPA experiences are associated with more negative affective and behavioral

outcomes when paired with high external regulation and lower levels of more autonomous forms of motivation.

In addition to testing psychological needs and behavioral regulations (i.e., motivation) as possible explanatory variables in the association between SPA experiences and PA, some researchers have tested constructs such as self-presentation and related efficacy perceptions (Cumming & Thøgersen-Ntoumani, 2011; Gammage, Hall, et al., 2004; Gammage, Martin Ginis, & Hall, 2004; Grieve, Jackson, Reece, Marklin, & Delaney, 2008; Strong, Martin Ginis, Mack, & Wilson, 2006; Woodgate et al., 2003). Individuals with higher levels of SPA report exercise motives to be more self-presentational (e.g., desire or need to present positive image of self; 'look good') compared to motives for fitness or recreation (e.g., enjoyment of activity, improvement of physical function; 'feel good') (Grieve et al., 2008; Strong et al., 2006). These findings are not surprising given the theoretical foundations of SPA linked to self-presentation. Furthermore, Cumming and Thøgersen-Ntoumani (2011) reported that self-presentational efficacy expectations moderated the relationship between SPA experiences and exercise frequency among adolescent females whereby SPA was negatively related to exercise frequency when self-presentational efficacy expectations were high. Similarly, in a sample of middle-aged and older adult women, Woodgate et al. (2003) found that self-presentational efficacy and SPA predicted little variance in PA. In this study, SPAS scores were negatively correlated with PA only for women with high self-presentational efficacy and unrelated to activity for females with low self-presentational efficacy. Finally, in a sample of male undergraduate students with weight training experience, Crozier (2012) attempted a self-presentational efficacy manipulation by assigning participants to complete a 1-Repetition Maximum (RM) test with either a muscular 'expert' trainer or a lean 'novice' trainer. The results of the manipulation were not significant, but participants assigned to the muscular trainer attained higher 1-RM scores and were perceived positively on musculature and expertise.

To date, most research on SPA and PA has been limited to a few cross-sectional studies or pre-post test manipulations with small samples. Overall, findings suggest higher reports of SPA in non-exercisers versus regular exercisers, and higher reports of SPA in athletes versus exercisers (Mack et al., 2008). Based on the literature reviewed, motivation regulations and self-presentational efficacy have shown to be important mediators in the SPA to PA association in leisure PA environments. No studies have examined associations as they relate to sport outcomes, such as drop-out, enjoyment, or performance, and there is limited work in the area of physical education. To advance the understanding of the associations between SPA experiences and PA outcomes, researchers need to expand on the identification of potential mediators and moderators that can help explain the often-reported ambiguous associations.

General discussion

The focus on SPA experiences has flourished since the inception of the concept in 1989. There are 126 research articles identified in this review in which researchers examine, test, identify, and define SPA experiences.

In fulfillment of the first objective, measurement issues prevalent in SPA research were reviewed and the fundamental properties of the SPA construct were examined. SPA has been measured using derivatives of the original 12-item scale (Hart et al., 1989), with most researchers turning to the original or a 9-item version of the SPAS (Martin et al., 1996).

Additionally, researchers report robust measurement properties of the 12-item, 9-item, and 7-item SPA scales across diverse samples, recommending the 7-item scale for more appropriate use in adolescents (Motl & Conroy, 2001). There is also evidence of sufficient measurement properties for a unique measure associated with social appearance anxiety (Hart et al., 2008) and a state-SPAS (Martin Ginis et al., 2011). This measure may be useful in experimental research on SPA due to its ability to discriminate between SPA experiences across various social and environmental contexts. This being said, the need for further scrutiny on the measure of SPA is ever-present. While researchers have provided some evidence of reliability and validity, prominent validity theorists (i.e., Messick, 1995) recognize that test/score validation is an ongoing process. Specifically, there are many sources of validity evidence (test content, internal structure, response processes, relations to other variables, consequences of test use, etc.) that should be the focus of further research on the SPAS and state-SPAS. Furthermore, although it is almost impossible to collect all the validity evidence at one time when developing and testing an instrument, it is recommended that multiple sources of validity evidence be collected simultaneously and presented (Zhu, 2012).

There has also been contestation in the literature about the nature of SPA experiences being trait or state-like. In longitudinal research, the stability of SPA is demonstrated over time and offers evidence of a stable disposition. However, the state-like properties of SPA are also demonstrated in experimental research which depicts the role of social context (e.g., presence of others) and environmental factors (e.g., presence of mirrors) in influencing state-SPA experiences. Based on the existing literature, it is likely that SPA has both trait and state features and needs to be measured in a way that is consistent with the objectives of the study.

Findings from this review also suggest that feelings of SPA may not be experienced frequently – a trend that is supported both in qualitative research among adolescents (McHugh et al., 2008; Sabiston et al., 2007) and by the consistent low scores on the SPAS scale in samples across the lifespan. It is possible that SPA is not easily self-identified (Sabiston et al., 2007) and that SPA is highly associated with a desire to maintain social desirability (Haase & Prapavessis, 1998). In fact, the difficulties in measuring emotions using survey instruments are well documented (Larsen & Fredrickson, 1999). While more objective assessments would be favorable, there are no known physiological or biological functioning responses that would be associated uniquely with self-presentational, social anxiety, and/or body image. Some researchers have linked body image and SPA-like outcomes to stress hormones (e.g., cortisol; Martin Ginis et al., 2012) and immune function (Sabiston, Castonguay, Barnett, O’Loughlin, & Lambert, 2009). It is possible that SPA experiences may have unique psychobiological profiles characterized by blunted stress hormone profiles and/or active systemic inflammation, although it is extremely premature to suggest this without more research.

To address the second objective, this review identifies the various predictors and outcomes of SPA that have been identified and tested within the literature. A preliminary overview of the complex associations between various factors and SPA is presented in Figure 1. Based on the evidence, personal characteristics (e.g., age, sex, physical self-perceptions) and social and environmental context variables (e.g., type of PA, characteristics of PA environment) have been identified as predictors. Personal characteristics (e.g., gender, age, maturation, ethnicity, weight status) have also been shown to moderate associations between predictors and SPA. Additionally, SPA has been associated with behavioral (e.g., coping strategies, motivation) and health-related (e.g., disordered eating,

stress, depression) outcomes. However, the associations in [Figure 1](#) are also premature in that they are based on predominantly cross-sectional research. As such, we urge careful use of this preliminary figure, and call for appreciation for the complex conceptualizations within the current SPA literature.

Finally, to address the third objective, this review aimed to explore the nomological networks that integrate SPA and PA behavior. Findings suggest that SPA has been studied most prominently as an antecedent to PA and related outcomes, and most studies report a negative association between SPAS scores and PA. Yet there is experimental evidence for a bi-directional relationship between SPA and PA, making it difficult to confidently make causal assumptions. The ambiguity of this relationship is also highlighted in a meta-analysis of SPA and PA (Mack et al., [2008](#)). An inherent roadblock lies in the inconsistent classification of individuals by PA level and extrapolation of results based predominantly on cross-sectional data (Mack et al., [2008](#)). The role of mechanisms such as motivation and self-presentational efficacy in the association between SPA and PA also needs to be further elucidated, as causal interpretations are still unclear. Experimental research is needed to further examine specific mechanisms explaining the association between SPA and PA.

Future directions

One of the challenges moving forward will be to determine the unique role of SPA within a comprehensive operationalization of body image affect, or body-related emotions. To better identify SPA experiences, it may be valuable to study other body-related emotions that may be more frequently experienced. One option is to turn some attention to self-conscious emotions (e.g., Sabiston et al., [2010](#)) that are informed by theoretical frameworks such as the cognitive-motivation-relational theory of emotion (Lazarus, [1991](#)) and the process model of self-conscious emotions (Tracy & Robins, [2004](#)). These frameworks may help to underpin the experience of emotions such as guilt, shame, pride, embarrassment, and envy, the distinction between emotions (e.g., shame versus guilt), and a focus on both negative and positive emotions. One of the limitations of studies focused on affect and emotion in sport and exercise psychology is the predominance of negative emotional experiences at the expense of also understanding positive emotions. A framework such as the study of body-related self-conscious emotions supports the understanding of a range of affective experiences.

Furthermore, the majority of research on SPA experiences has focused on females and either adolescents or undergraduate/college students. While there are 45 studies (35.7%) exploring gender differences more broadly, there are only a few ($n = 7$; 5.6%) studies focused specifically on males. Also, there are limited ($n = 12$; 9.5%) research studies targeting potentially important samples such as individuals of diverse race/ethnicities, middle-aged women, older adults, and clinical populations. Researchers need to consider branching out to diverse samples to best understand SPA experiences, and to consider doing so in the context of longitudinal designs. Longitudinal studies will provide insights into SPA across the developmental lifespan to better understand change in SPA experiences in relation to other relevant variables, such as PA behavior.

Considering evidence that SPA is often negatively linked with PA adherence (Cox et al., [2011](#); Treasure et al., [1998](#)) and participation (Brunet & Sabiston, [2009](#); Kowalski et al., [2001](#); Lantz et al., [1997](#)), there is value in conducting clinical intervention studies to target ways of reducing SPA experiences and fostering more self-determined motivations for PA.

To date, there is virtually no clinical intervention research on SPA. Drawing from the broader domains of body image research, some intervention strategies may be valuable approaches to use in sport and exercise psychology practice. For example, interventions based on psychodynamic approaches (Krueger, 2002), experiential psychotherapy and personal growth (Rabinor & Bilich, 2002), and cognitive-behavioral therapy (Cash & Strachan, 2002) have been used with evidence of reduced body image concerns, with the latter approaches demonstrating the most effective outcomes. Specifically, cognitive-behavioral therapies include psycho-education, biofeedback, desensitization, and cognitive dissonance training, which may be successfully utilized in sport, exercise, and clinical settings. Further strategies may include targeting self-presentational efficacy.

The need for intervention research is likely targeted at specific sub-groups in which the association between SPA and PA is negative. Identifying these sub-groups is another avenue for future research. Furthermore, it would be equally imperative to better understand the contexts in which the association between SPA and PA is positive (Aşçi et al., 2006; Bowden et al., 2005; Frederick & Morrison, 1996; Hausenblas et al., 2004). While some researchers have turned to self-determination theory (Deci & Ryan, 2002) and stress and coping perspectives (Kowalski et al., 2006; Lamarche, Kerr, Faulkner, Gammage, & Klentrou, 2012; Niefer, McDonough, & Kowalski, 2010; Sabiston et al., 2007) to explain the potential positive relationship, other theoretical and conceptual models are likely of value and need to be identified and tested.

Most of the research on SPA experiences has focused on PA settings such as sport or exercise, whereas SPA experiences in physical education have received minimal attention (Cox, Ullrich-French, & Sabiston, 2013; Cox et al., 2011; Koca & Aşçi, 2006). This is surprising given the evaluative and social nature of this environment (Crombie, Brunet, & Sabiston, 2011) and the vulnerability of the adolescent participants who place significant emphasis on appearance and social relationships (Harter, 2012). Examining SPA in specific PA settings such as physical education may also lead to the development of context-specific strategies for reducing SPA experiences. For example, Cox et al. (2011) showed support for peer acceptance and autonomous motivation possibly being protective against SPA experiences in physical education contexts. In addition, Crombie et al. (2011) have suggested that increasing self-efficacy, increasing perceptions of control, and fostering acceptance within physical education may help decrease body-related concerns such as SPA. Despite these specific recommendations, there is currently no evidence on the development or implementation of intervention strategies to target these mechanisms and decrease the burden of SPA experiences.

Additionally, there are a multitude of dynamic relationships in the physical education and sport contexts that remain unexplored. For example, it may be useful for future research to examine the role of physical educators and coaches in transmitting beliefs of physical self-perceptions and performance to students and athletes, which may help us to better understand experiences of SPA. In fact, Koca and Aşçi (2006) suggest that future studies consider how variables such as curriculum content and instructional methods may influence experiences of SPA in youth in physical education contexts. Taken together, these suggestions for future research can inform intervention strategies for youth, coaches, and educators.

Finally, one of the biggest limitations in the SPA literature to date is the lack of longitudinal studies. Prospective studies are needed to provide insight on the developmental trends in SPA experiences, health outcomes, performance outcomes, and an understanding of

the mechanisms that may help explain the association between SPA and behavioral outcomes among exercisers, non-exercisers, and athletes.

Concluding remarks

This review is intended to be a broad and comprehensive summary of the research in this area to date. The purpose of this review was to present a summary of the SPA research in sport and exercise psychology. Based on the synthesis, SPA experiences show variation across PA context, sex, age, and culture. Activity motives are important mechanisms linking SPA and PA outcomes in exercise, sport, and physical education. Researchers in sport and exercise psychology are now encouraged to broaden the study of SPA to conduct longitudinal and experimental studies, develop quality intervention strategies aimed at reducing SPAS scores, and expand on the theoretical, nomological, and operational understandings of SPA experiences across the lifespan.

Supplemental data

Supplemental Tables 1 and 2 can be accessed at <http://dx.doi.org/10.1080/1750984X.2014.904392>. The underlying research materials for this article can be accessed at <http://physical.utoronto.ca/health-and-exercise-psychology-unit/publications/peer-reviewed-research-articles>.

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